BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Benchmark research release 2.1
Population model and human studies pending

Abstract

Visual assembly planning requires predicting how an action changes later access, workspace occupancy, and unfinished work. Final-shape accuracy and solution length alone do not isolate these abilities. We introduce BrickPlan, an executable benchmark with exact future-cost and recovery labels under a declared, finite assembly abstraction. Original LDraw geometry determines qualified motions and connectivity; a limited rack and persistent off-assembly inspection ledger couple decisions over time. The release contains 145 interactive episodes and 739 diagnostic questions from 11 source groups. Matched sequence pairs balance early and late conflicts, geometry contrasts change future legality while preserving registered pieces and plans, and delayed-repair questions exclude the terminal goal from recovery credit. We release native-resolution observations, a categorized question explorer, exact evaluators, and a real four-call local VLM trace. Structural controls reveal both scope and limits: closure greedy completes 384 of 435 task-runs but only 266 optimally, while full closure A* solves every run with at most 4,372 expansions. A five-step prefix shortcut scores 66.7% on the sequence tests. These results validate measurable scheduling and diagnostic distinctions, not necessary internal reasoning depth. Population VLM and human results remain unmeasured.

1. Introduction

An assembly action can be legal now and expensive later. Removing a part may consume the last available holder before a deeper part is accessible; reattaching another part may block unfinished work. If a planner notices an error only after several more operations, undoing its latest action need not restore an efficient completion. Measuring these effects requires more than recognizing brick types or comparing a final image.

BrickPlan asks a specific question: can a visual agent predict the future action cost of a decision, and recover after that decision has propagated? We separate three observable failures: an immediately rejected operation, a legal action that increases exact remaining cost, and a later failure to return to an optimal completion corridor. The geometry and tasks are small enough to label these distinctions exhaustively. This deliberate restriction enables exact diagnostics at the expense of unrestricted manipulation.

A motivating audit illustrates why such care is necessary. For a preceding assembly-only bank, an occupied-blocker closure formula reproduced all 1,802,728 stored state–goal costs. Long solutions in that bank did not establish deep planning. Our task adds limited external storage and inspection work that persists after reattachment. Static geometric relations remain useful, but physical occupancy alone no longer specifies the mission state or its exact cost.

We make three contributions. First, we define a visual assembly task with certified, state-dependent resource and work obligations, accompanied by both interactive episodes and explicit diagnostic question formats. Second, we construct sequence, geometry, appearance, and delayed-repair controls that expose particular shortcuts and confounds. Third, we release the observation, execution, scoring, and publication pipeline together, including exact software controls and an authenticated model integration case. We distinguish these contributions from empirical claims that await a larger model and human study.

2. Evaluation design

Table 1 fixes the reporting structure before empirical collection. Primary outcomes are assigned-episode completion, optimal completion, and excess operations among successes. Diagnostic accuracies are reported by question family; pair scores require both members to be correct. Invalid responses and missing assignments remain visible in denominators. Software controls with exact relations are separated from the public image and structured conditions.

The proposed study compares matched source

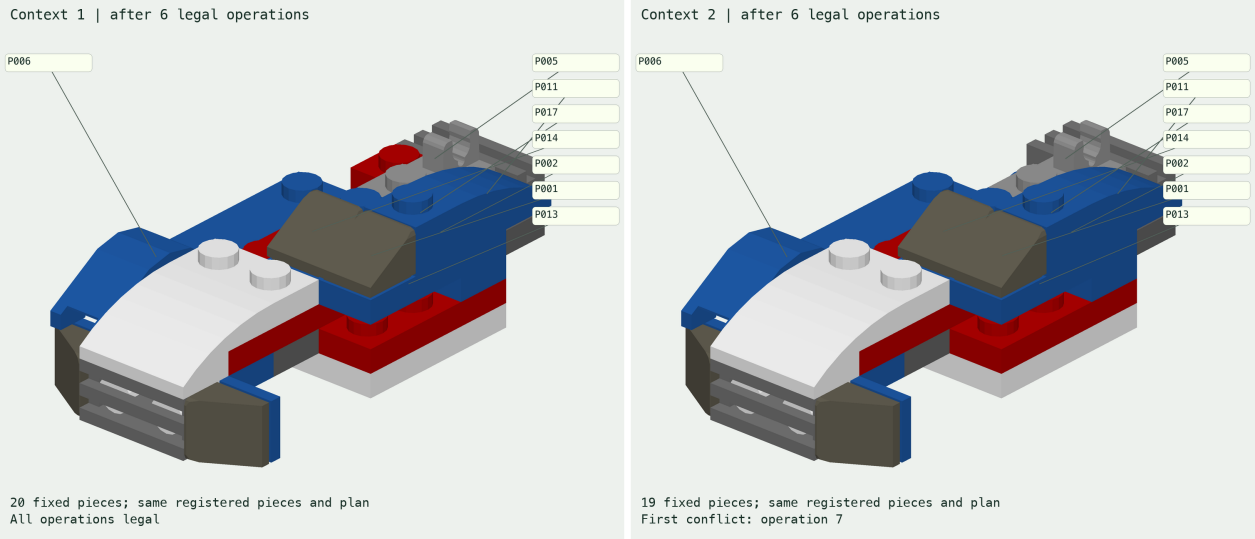


Figure 1. An actual answer-changing geometry pair after its shared six-operation legal prefix. Registered pieces, original poses, initial occupancy, goal, rack, work orders, and proposed plan are matched. The fixed context differs by one original source piece. Context 1 permits the whole sequence; context 2 rejects operation 7. These are qualified induced source assemblies, not edited relation labels. The unequal fixed-part counts remain a confound.

Table 1. Experimental conditions. "Pending" means no population result has been collected; the local trace is a single integration case. Model snapshots and provider settings must be fixed before the study.

Condition	Evidence status
Structural policies and exact search	Measured
SmolVLM-256M, image input	One failed case
Open VLMs, image / structured	Pending
Proprietary VLMs, image / structured	Pending
Humans, image + registered IDs	Pending
Stepwise / explicit-plan prompting	Pending

groups and identity seeds under equal action and observation budgets. For each VLM, both input conditions use the same executor and answer syntax. A format/cost pilot should precede the full study, with pilot results retained and identified. Report model snapshot, provider, prompt, image preprocessing, context policy, decoding, token usage, and calls per assignment. Human collection requires recorded instructions, training exposure, participant counts, and consent procedures before recruitment. No such cohort is claimed here.

Repeated identity seeds and paired questions are correlated observations. Report per-source results and coverage before a pooled mean. With only three newly sampled source groups, uncertainty for transfer is necessarily limited; a large number of questions does not create a large number of independent assemblies. Stratify errors into parsing, immediate legality, future cost, diagnosis, and repair so a format failure is not interpreted as evidence of deep planning failure.

3. Related work

LTRON's Break and Make evaluates interactive structural understanding through disassembly and reconstruction [5]. InstructioNet studies construction supported by an agent's own explicit visual instructions [6]. These establish interactive brick construction and long sequences; BrickPlan does not claim either as new. Its narrower registered-site action space permits exact counterfactual completion costs and event-level recovery labels.

LEGO-Puzzles organizes spatial understanding and sequential reasoning into explicit task families [4]. Its revised version includes both elementary VQA and a Planning set; it should not be reduced to a static recognition benchmark. PhyBlock combines hierarchical assembly and physical-understanding questions [3]. Brick-Composer introduces BC-Bench for diverse brick selection and pose estimation, and trains models with construction and feedback signals [2]. BrickPlan instead fixes legal placement sites and asks about resource coupling, future legality, and delayed recovery. It complements these broader spatial tasks rather than replacing them.

Shortest-path search and admissible heuristics are established planning tools [1]. A failed weak policy

does not establish computational hardness. We therefore report a strong static relaxation, resource-aware dispatch, bounded and full A*, and optimistic shallow-choice bounds. The research contribution is an audited visual diagnostic instrument; finite racks or A* are not algorithmic novelties.

4. Task and observation contract

4.1. State, geometry, and actions

Let $A \subseteq V$ be occupied registered sites, $G \subseteq V$ the target, $J \subseteq V$ the required inspection sites, and $D \subseteq J$ the completed inspections. The mission state is $x = (A, D)$. Each physical piece P_i is registered to exactly one original site S_i. Detached pieces occupy universal oversized rack holders:

$$r(x) = |V \setminus A| \leq K, \tag{1}$$

where K is the declared capacity. An audit completes one pending job only when its piece is detached; completion persists after reattachment. The terminal physical condition is $A = G,\ D = J$.

An attach or detach must satisfy source-qualified straight translation, collision, stud-connectivity, registration, and capacity rules. Fixed pieces form a connected clamped assembly. All executable positions are original LDraw poses. Qualification uses guarded inset meshes and prescribed paths; it is not force, gravity, clutch, hand-access, or arbitrary rigid-body simulation. The rack is an explicit resource abstraction, not simulated transfer geometry. Exploded displays are marked reference diagrams and never executable states.

Each accepted physical operation or audit costs one operation, as does each rejected operation attempt. Valid camera acquisition and diagnosis consume their separate budgets; finish explicitly terminates a completed task. There are 12 additional views and 12 diagnoses per episode. The action budget is fixed in the released task record before collection. Source size, work orders, budget, and rack capacity are public constraints.

4.2. Matched observation conditions

The image condition provides five PNG panels: current assembly, target, all-site reference, registered-piece catalog, and rack. Main views are rendered at 3840×2880 and antialiased to 1920×1440; individual catalog pieces have 480×360 native tiles. Public P/S registration and the inspection ledger are textual. Image input omits pose matrices, installed-site flags, legal-action lists, relation graphs, and exact costs. These omissions preserve a visual inference component, while explicit registration and rack inventory simplify correspondence.

The structured condition adds original matrices, part identifiers, colors, path directions and lengths, and site occupancy. It still omits the private blocker/support graph. Structural software controls receive that graph and are a separate privileged condition. Additional acquired views persist as revision-specific observations. Free 3D orbit and explosion are available in the unscored practice workbench, outside study and model observations. Human and model policies share the controlled observation and submission protocol.

4.3. Explicit diagnostic formats

Table 2 distinguishes interaction from frozen-checkpoint questions. The 230 next-action questions expose only legal candidates; any candidate minimizing exact completion cost is correct. The 144 continuation questions ask for the first illegal operation in a fully specified sequence, with null denoting complete legality. Eight geometry and 24 appearance questions use the same answer format. The 333 delayed-repair questions include the pre-error checkpoint, executed history, current observation, and legal candidates. They ask for the next action on a shortest repair path, not a model's hidden reasoning.

5. Construction and validity controls

5.1. Source grouping and sampling

The release contains 20 qualified geometry profiles from 11 source kit groups. This profile count includes geometry candidates that yield no retained tasks. There are 109 development episodes from eight groups and 36 newly sampled episodes from three groups: 10014, 10128, and 31031. Alternate builds and geometry variants remain within their source group. Roadster and kart, for example, share kit 31027 and are never treated as independent sources.

Legacy development tasks were selected using mechanism and solver evidence. For additional sources, the candidate list and seeded sampling plan were recorded before new-source solver evaluation. Initial geometric qualification rejected source collisions. We then refined admission to remove conflicting source instances in deterministic source order and retain a stud-connected component. This is a disclosed geometry-only refinement, not an unchanged preregistered algorithm. It moves no bricks. After qualification, task admission uses seeded sampling and solvability, not measured policy difficulty. Easy new-source cases are retained. The initial failures, retained source indices, and exact qualification records are released.

"New source" refers to benchmark construction, not

Table 2. Seven question families in the public explorer. Each question has a stable ID, complete prompt, state, constraints, answer format, and scoring rule. Episode families are mutually exclusive presentation groups.

Family	Question	Response and measurement
Assembly / reconfiguration	Transform the current assembly into the target.	Action sequence; completion and excess cost
Finite-resource scheduling	Which operations free the holders needed for later work?	Action sequence under a capacity bound
Inspection / persistent state	Complete off-assembly work and restore the target.	Actions plus audits; target and ledger completion
Next-action judgment	Which legal choice minimizes total completion cost?	Candidate index; any optimal candidate accepted
Future-sequence checking	Where does this complete proposed plan first become illegal?	One-based conflict index, or all legal
Visual geometry / appearance	Does a support change alter the answer? Does recoloring preserve it?	First-conflict answer and joint pair correctness
Delayed error recovery	After an error and more actions, which choice repairs most efficiently?	Candidate on a shortest nonterminal corridor-return path

Table 3. Operation-length strata and measured descriptors. D1–D4 are predetermined length bins (1–8, 9–16, 17–24, 25+), not cognitive-depth levels. Gap is exact cost minus the static lower bound.

Tier	Episodes	C^* range	Gap range	A* max
D1	16	1–8	0–0	8
D2	47	9–16	0–2	58
D3	46	17–24	0–8	4372
D4	36	25–42	0–18	2555

absence from model pretraining. The partition is public. The dataset does not support a claim of contamination-free evaluation.

5.2. Difficulty descriptors and exact labels

D1–D4 denote optimal-operation-length bins only (Table 3). We retain separate descriptors for registered pieces, capacity, initial occupancy, work orders, heuristic gap, and observed search effort. No new tier is assigned using an A* threshold. An optimum's length remains a property of the shortest output sequence, not proof of required internal model depth.

For each mission, we enumerate the capacity-constrained physical component and its inspection-bit product. Reverse directed shortest paths yield $C^*(x)$, the minimum remaining operation count. All 70 oracle tables are checked against runtime transitions, physical inverses, lower bounds, and Bellman equalities. Across tables there are 4,053,285 product-state entries and 39,989,367 directed-transition entries. These totals repeat physical states under different goals and jobs.

Let R be the occupied blocker closure needed to remove unwanted pieces, insert missing target pieces, and detach unfinished inspection pieces. A strong static relaxation is

$$h(x) = |A \triangle G| + 2|R \cap A \cap G| + |J \setminus D|. \qquad (2)$$

Each mismatch needs an operation; each occupied target piece in the required removal closure must be removed and replaced; each unfinished audit costs one. Capacity and connectivity may add operations, preserving admissibility. The maximum retained gap $C^* - h$ is 18.

5.3. Future-sequence shortcut controls

We construct 72 legal/corrupted pairs with identical first and last actions, identical action multisets, and identical legal prefixes before the corrupted sequence's first conflict. Reordering changes intermediate validity. The legal sequence contains no repeated states or no-op padding. There are 24 pairs in each first-conflict bin: positions 2–5, 6–10, and 11 or later.

A prefix-only control executes the first k operations and otherwise answers "all legal." It scores 80/144 at $k = 2$ and 96/144 at $k = 5$, versus 73/98 and 92/98 in frozen version 2.0. Balanced labels give the zero-step control 50%. The revision reduces an observed shortcut; it does not establish immunity to other ones. All 230 selected next-action checkpoints still include an optimal choice in the depth-one closure heuristic's minimizing set. Ambiguous ties, rather than exclusion of every correct shallow choice, are the relevant measured phenomenon.

5.4. Geometry and appearance contrasts

Four retained geometry pairs change only the induced fixed support context while preserving registered pieces, original poses, initial occupancy, goal, capacity, work orders, and proposed sequence. Both contexts are independently qualified for meshes and motion paths. Their first answer-changing conflict follows a shared legal prefix (Figure 1). This links the question to geometric context beyond a text-only capacity rule.

The scope is narrow: all four pairs come from one development source, and their fixed-part counts differ. Equal-count/type contexts from three other base profiles were qualified but yielded no retained answer-changing sequence. We publish that zero-yield census and do not call the retained pairs count-matched visual counterfactuals. Twelve additional pairs use original-color and grayscale images of exactly the same state and plan. They measure answer invariance to appearance; they are correlated diagnostic controls, not new independent structures.

5.5. Delayed repair

For pre-error state b, let $\mathcal{C}_b^-$ be the union of nonterminal states on all optimal completions from b. After a legal positive-excess error, the generator executes a distinct-state legal history of lag 1, 3, or 5 that remains outside this corridor. At delayed checkpoint z, exact repair cost is

$$r_b(z) = \min_{y \in \mathcal{C}_b^-} \operatorname{dist}(z, y). \tag{3}$$

The 333 questions have minimum repair costs from 1 to 6, verified by independent forward BFS. A candidate is correct if it lies on a shortest repair path. The goal is excluded: finishing after a costly detour alone does not demonstrate that the agent restored an efficient earlier plan. The visible checkpoint and history define the reference corridor unambiguously.

6. Metrics

For a legal unit-cost action a with successor x', conditional excess is

$$\delta(x, a) = 1 + C^*(x') - C^*(x) \geq 0. \tag{4}$$

A rejected operation leaves state unchanged and contributes one to total excess. Thus legal but costly actions are distinguished from immediate invalidity, including when the checkpoint already requires rework. For a successful trace of T charged attempts, regret is $T - C^*(x_0)$. Primary reporting includes completion and optimal completion over all assigned episodes; successful mean regret is accompanied by its success coverage.

Choice diagnostics distinguish avoidable-cost decisions, loss of remaining-budget viability, and genuine unreachability. Each choice rate uses checkpoints where the corresponding good and bad legal alternatives both exist. The retained physical components are reversible, and completed audits never remove physical actions. Consequently, genuine irreversible dead ends are absent and their eligible-choice rate is undefined.

Every legal positive-excess action opens a separate recovery event. Later errors do not overwrite earlier events. The evaluator records immediate return and delayed cohorts at lags 1/3/5, detects corridor restoration, and preserves right-censored unfinished trajectories. Diagnosis asks whether an avoidable legal error occurred since the last diagnosis; mandatory rework alone is not an error. Replanning reports compare actual suffix effort with the exact remaining cost at the diagnosis checkpoint.

Plan-survival curves preserve the first cost-loss event even if recovery later occurs. They report the assigned-task denominator at each operation index and distinguish missing traces from known outcomes. Joint pair correctness requires both assigned members; a missing member earns no joint credit. Supplementary definitions specify normalization, censoring, and the audit-inclusive regret conservation identity.

7. Measured evidence

7.1. Structural policies

Closure greedy minimizes the static successor bound with visited-state and randomized site tie breaking. Capacity dispatch prioritizes audits, mandatory removals, safe placements, and temporary staging, with escape from revisited states. Both are competent no-search controls within this abstraction. Greedy completes 384/435 runs, dispatch 367/435, and each achieves 266 optimal completions. Completion alone therefore conceals substantial avoidable cost (Table 4).

A* uses graph reopening, $(g + h, h)$ priority, and declared expansion limits without consulting oracle tables. Caps of 100 and 1,000 expansions complete 362 and 404 runs, respectively. Full closure A* completes all 435 optimally, with median 20 and maximum 4,372 expansions. The weaker occupancy-plus-pending-audit heuristic also solves all runs, but uses up to 75,454 expansions. This contrast demonstrates how a weak baseline can inflate apparent difficulty. The full-information problem remains computationally modest.

Dropping inspection obligations completes only 156 actual missions, although its relaxed target is reached in every run. Ignoring the rack limit completes 239, with 196 executor rejections. These deliber-

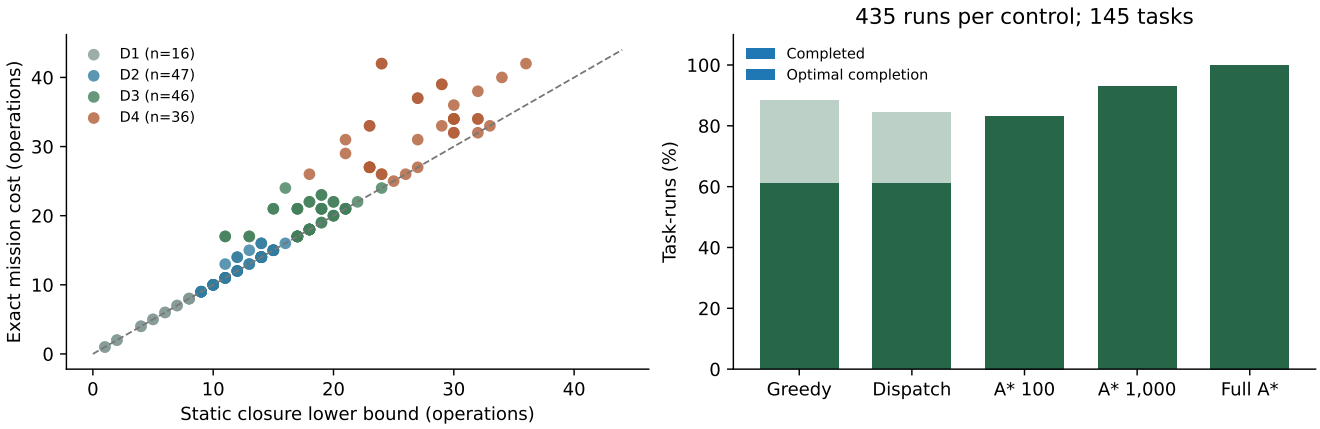


Figure 2. Measured task and control evidence. Left: exact cost exceeds the static bound on some missions; coincident points overlap. Right: successful completion and optimal completion differ for greedy policies. Length strata do not impose a monotonic ordering of search difficulty. These descriptive measurements mix legacy development selection and new-source seeded sampling, whose results should also be examined separately.

Table 4. Structural controls on 145 episodes. The oracle uses one seed; all other policies use three tie-order seeds (435 runs). Regret is mean excess among successful runs. Relations are privileged, so these are not image-track VLM results.

Control	Runs	Success	Optimal	Regret	Max exp.
Exact oracle	145	145	145	0.00	—
Closure greedy	435	384	266	7.29	—
Capacity dispatch	435	367	266	7.44	—
Closure A* (100)	435	362	362	0.00	100
Closure A* (1,000)	435	404	404	0.00	1,000
Closure A* (full)	435	435	435	0.00	4,372
Hamming A* (full)	435	435	435	0.00	75,454
Legal random	435	25	10	9.76	—
Drop inspections	435	156	156	0.00	—
Ignore rack limit	435	239	239	0.00	—

ate model-mismatch ablations diagnose omitted constraints rather than representing strong planners. Raw actions, final states, rejection codes, expansions, timings, and every aggregate are released. Repeated seeds reuse source geometry, and local cached timings are not calibrated cross-platform comparisons.

7.2. Real model integration case

The default explorer case runs SmolVLM-256M-Instruct on kart-deck-08 (public E069, seed 0). It is the first local run on that demonstration, with all four calls retained. Each call returns a scene description instead of a JSON action; each becomes INVALID_ACTION, consumes one operation, and leaves the physical state unchanged. The controller then stops at the four-call limit. This failed case verifies the actual model–feedback loop, not planning competence or population performance.

The archive includes exact public prompts and observations, image hashes, raw responses, parsed actions, feedback, before/after frames, latency, measured MLX token counts, and the weight digest. Native stimuli remain 1920 pixels wide; this local model's processor resizes the longest edge to 512 with image splitting disabled. That preprocessing is explicitly recorded. An OpenAI-compatible vision adapter supports further practice runs with server-side credentials. Population experiments use a separate usage-accounted study runner.

8. Release, limitations, and outlook

The explorer starts with the seven task families, complete question previews, and filters for stable ID, source, split, and length bin. Users can attempt a question, inspect submitted-answer feedback, watch model calls, and export review JSON. A React workbench and isolated WebGL2 renderer retain original source geometry

in live practice. Native zoom preserves raster evidence. Dataset cards, public JSONL, sample HD observations, evaluator dependencies, and deployment materials are packaged together. They are local release artifacts; an external hosting URL is not claimed.

Software validation replays all 145 optimal episodes and 739 diagnostic answers. Additional audits check matched prefixes/action multisets, source-group separation, repair BFS labels, budget accounting, overlapping errors, missing assignments, and altered transcripts. Protocol and browser tests cover persistence, idempotence, observation isolation, question clarity, HD zoom, and model feedback. This establishes software behavior, not human usability or hardware validity.

The principal remaining limits are empirical and structural. The public bank is small, deterministic, and tractable with perfect relations. Registered sites and explicit inventories simplify visual grounding. The visual-context evidence is limited to one source and a part-count confound. Geometry qualification omits forces, free poses, rotation, simultaneous subassembly motion, and real rack transfers. No model shortsightedness, human–model gap, or robustness to all shortcuts has been measured. The next research step is the preregistered matched image/structured study, with independent source replication and additional count-matched geometric contrasts. BrickPlan provides the executable instrument and falsifiable measurements for that study.

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